

Effects of naltrexone on food intake and changes in subjective appetite during eating: evidence for opioid involvement in the appetizer effect.

The effects of 50 mg naltrexone on eating and subjective appetite were assessed in a double-blind placebo-controlled study with 20 male volunteers. Appetite was monitored using a disguised digital balance connected to a micro-computer, which constantly monitored the amount of food remaining, and which automatically interrupted feeding for 30 s after every 50 g consumed to allow appetite ratings to be made. Half the subjects ate pasta with a cheese sauce, and the remainder pasta with a tomato sauce. Subjects ate significantly less of both foods after 50 mg naltrexone than in either the placebo condition or on the initial (familiarisation) day. Naltrexone also reduced the rated pleasantness of both foods, and reduced overall eating rate. When best-fit quadratic functions were used to describe changes in rated hunger in relation to intake within the meal, naltrexone abolished the positive linear component reflecting the initial stimulation of appetite without altering either intercept or the negative quadratic function. Although mood ratings suggested that naltrexone had a mild sedative effect, mood changes alone could not explain the effects of naltrexone on appetite. Overall, these data suggest a specific role for opioids in the stimulation of appetite through palatability.